

Ajay Rajendra Kumar

RESEARCH ASSISTANT · VISION-LANGUAGE MODELS

Boston, MA, USA

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“Towards Generalizable and Reliable Artificial Intelligence”

Summary

Computer science graduate conducting research in computer vision and multimodal learning using large multimodal models (MLLMs). Focused on video understanding with vision-language models (VLMs), improving spatial and temporal reasoning. Working on visual grounding strategies using agentic architectures and expert tools to improve performance in visual question answering. Research assistant in Prof. Lorenzo Torresani's lab at Northeastern University; previously part of the Visual Intelligence Lab at Northeastern.

Highlights

- Co-authored a paper under review at IEEE IROS 2026, with a demo accepted at CVPR 2026 and a workshop paper accepted at ICRA 2026.
- Academic honors & awards:
 - Most Innovative Project among 40+ submissions, PhD-level Deep Learning Course, Northeastern University (Mar. 2025).
 - National 2nd Runner-Up, Daimler India Commercial Vehicles Hackathon (Aug. 2022).
 - 1st Runner-Up, Transfusion Medicine Hackathon (Apr. 2022).
 - Winner, Space Odyssey Web Development Hackathon (Jun. 2021).
 - Scholar's Scholarship Award, Manipal Academy of Higher Education (Jan. 2021).
- Research experience across computer vision, NLP, and robotics at Northeastern University (Prof. Lorenzo Torresani, Prof. Huaizu Jiang), IIT Kharagpur, Institute for Plasma Research, and Bhabha Atomic Research Center.
- Industry experience as a Software Development Intern at Fidelity Investments; received a full-time return offer.

Field of Interest

MY RESEARCH INTERESTS INCLUDE, BUT ARE NOT CONFINED TO:

- Video understanding and spatiotemporal reasoning in vision-language models
- Visual question answering and multimodal reasoning
- Efficient deep learning for real-time perception
- Dense visual SLAM and 3D scene reconstruction on edge devices

Education

Northeastern University

Boston, USA

M.S. IN COMPUTER SCIENCE

Sep. 2024 - May 2026

- GPA: 4.0/4.0
- Dissertation: “Dense Visual SLAM on Edge Devices”
- Advised by Prof. Huaizu Jiang

Manipal Institute of Technology

Manipal, India

B.TECH. IN COMPUTER SCIENCE

Aug. 2020 - Jul. 2024

- GPA: 9.3/10.0
- Minor: Computational Intelligence
- Dissertation: “Evaluating Deep Learning Approaches for Fracture Segmentation in X-ray Images”
- Advised by Prof. Subhamoy Mandal

Work Experience

Research Assistant

Boston, USA

NORTHEASTERN UNIVERSITY, KHOURY COLLEGE OF COMPUTER SCIENCES

Nov. 2025 - Present

- Serving as a research intern at Prof. Lorenzo Torresani's lab.
- Working on improving spatiotemporal reasoning in visual question answering tasks for vision-language models.

Research Assistant

Boston, USA

NORTHEASTERN UNIVERSITY, KHOURY COLLEGE OF COMPUTER SCIENCES

Apr. 2025 - Mar. 2026

- Served as a research intern at Prof. Huaizu Jiang's lab.
- Worked on dense visual-SLAM system capable of running real-time on edge devices.
- Submitted a paper to IEEE IROS 2026 (under review).
- Demo accepted at CVPR 2026; workshop paper accepted at ICRA S2S Workshop 2026.

Research Assistant

NORTHEASTERN UNIVERSITY, D'AMORE-MCKIM SCHOOL OF BUSINESS

Boston, USA
Feb. 2025 - May 2026

- Served as a graduate research assistant at Prof. Anna Lamin's lab.
- Contributed in projects to evaluate CSR initiatives driving social impact and sustainable development outcomes in India.
- Collaborated with Indian Institute of Management Bangalore on analyzing the impact of corporate growth in Bengaluru on housing.

Undergraduate Researcher (Bachelor's Thesis)

INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR

Kharagpur, India
Jan. 2024 - Jul. 2024

- Advised by Prof. Subhamoy Mandal.
- Worked on semantic segmentation of humerus bone from X-Ray images.

Software Development Intern

FIDELITY INVESTMENTS

Bengaluru, India
Jun. 2023 - Aug. 2023

- Redesigned change ticket management system using Angular and Spring Boot.
- Obtained a full-time offer to join Fidelity.

Project Intern

INSTITUTE FOR PLASMA RESEARCH

Gandhinagar, India
Apr. 2023 - Dec. 2023

- Advised by Prof. Rajaraman Ganesh.
- Worked on simulation of turbulent fluid flows.
- Delivered experimental simulation software to the computing division.

Project Intern

BHABHA ATOMIC RESEARCH CENTER

Visakhapatnam, India
Dec. 2022 - Jan. 2023

- Advised by Dr. Manoj Warriar.
- Contributed to early-stage research on modeling spectral neighbor analysis potentials for Zirconium.

Selected Projects

Pose Graph Sparsification by Maximizing Algebraic Connectivity

Northeastern University, Boston,
USA
Mar. 2025

MOBILE ROBOTICS

- Implemented algebraic connectivity maximization for pose graph sparsification.
- Evaluated on benchmark pose graphs using SE-Sync, achieving sparser graphs with minimal performance loss.
- Keywords: SLAM, Pose Graph Sparsification, Python

Hallucination Detection in Large Language Models

Northeastern University, Boston,
USA
Mar. 2025

NATURAL LANGUAGE PROCESSING

- Evaluated hallucination detection methods from "Joint Evaluation of Answer and Reasoning Consistency for Hallucination Detection in Large Reasoning Models" (AAAI 2026) across multiple open-source LLMs.
- Identified limited generalization of the proposed methods beyond the original evaluated models.
- Keywords: Large Language Models, Hallucination Detection, PyTorch, Python

Improving Text-Driven Human Motion Generation through Equivariance

Northeastern University, Boston,
USA
Mar. 2025

DEEP LEARNING

- Integrated equivariant neural networks into motion diffusion models for human motion generation.
- Improved matching accuracy and achieved competitive FID scores against existing baselines.
- Keywords: Human Motion Generation, Diffusion Models, Equivariance, PyTorch

Rethinking Depth for Object Detection with YOLOv3

Northeastern University, Boston,
USA
Dec. 2024

COMPUTER VISION

- Implemented YOLOv3 and RGB-D variants to study the impact of depth cues on object detection.
- Observed limited performance gains, indicating need for improved depth fusion architectures.
- Keywords: Object Detection, RGB-D, YOLOv3, PyTorch

Addressing Vaccine Misinformation on Social Media

Manipal Institute of Technology,
Manipal, India
Mar. 2024

RESEARCH ASSISTANT (BACHELOR'S)

- Fine-tuned transformer models for misinformation classification on social media data.
- Developed similarity-based method for anomalous user detection, achieving strong classification performance.
- Keywords: Misinformation Detection, Transformers, NLP, Python

Publications

Underline denotes co-first authors or co-corresponding authors

INTERNATIONAL CONFERENCE

1. Aniket Gupta, Tianye Ding, **Ajay Rajendra Kumar**, Dennis Giaya, Pia Bideau, Charles Saunders, Qu Cao, Aruni RoyChowdhury, Hanumant Singh, and Huaizu Jiang, “NeuSLAM: Dense Visual SLAM on Edge Devices,” under review at *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, 2026; accepted to *ICRA 2026 Workshop on From Sea to Space: Advancing Perception in Harsh Domains*; accepted to *IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) 2026 Demo Track*.
2. C. Rao, G. M. Prabhu, **Ajay Rajendra Kumar**, S. Gupta, and N. P. Shetty, “Addressing Vaccine Misinformation on Social Media by Leveraging Transformers and User Association Dynamics,” in *Int. Conf. Mach. Learn. Data Eng. (ICMLDE)*, 2023.

Teaching Experience

Object-Oriented Design Lab (CS2163)

UNDERGRADUATE TEACHING ASSISTANT (INSTRUCTOR: PROF. ANUP BHAT)

*Manipal Institute of Technology,
Manipal, India*

Aug. 2023 - Dec. 2023

Problem Solving Using Computers Lab (CS1081)

UNDERGRADUATE TEACHING ASSISTANT (INSTRUCTOR: PROF. T. SUJITHRA)

*Manipal Institute of Technology,
Manipal, India*

Aug. 2023 - Dec. 2023

Honors & Awards

Mar. 2025 **Most Innovative Project (among 40+ submissions)**, PhD-level Deep Learning Course,
Northeastern University

Boston, USA

Oct. 2023 **Undergraduate Sponsorship Grant**, Manipal Academy of Higher Education (MAHE)

Dehradun, India

Aug. 2022 **National 2nd Runner-Up**, Daimler India Commercial Vehicles Hackathon

India

Apr. 2022 **1st Runner-Up**, Transfusion Medicine Hackathon

Manipal, India

Jun. 2021 **Winner**, Space Odyssey Web Development Hackathon

Manipal, India

Jan. 2021 **Scholar's Scholarship Award**, Manipal Academy of Higher Education (MAHE)

Manipal, India